

MTH 446: Lecture # 3 (catching up)

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Lecture Span

1. Properties of Argument

Let $z \in \mathbb{C}$, $z \neq 0$, $\varphi \in \arg(z)$, and if $-\pi < \varphi \leq \pi$, then φ is called the principal value of $\arg(z)$ with notation

$$\varphi = \arg(z) \tag{1}$$

Therefore,

$$\varphi = \arg(z) \iff \varphi \in \arg(z) \wedge -\pi < \varphi \leq \pi \tag{2}$$

As an example with

$$\arg(i) = \{\pi/2 + 2\pi n, n \in \mathbb{Z}\} \tag{3}$$

Then, use $\pi/2 \in \arg(z)$ as the principal value. Thus,

$$\arg(i) = \pi/2 \tag{4}$$

Also,

$$\arg(-1) = \{\pi + 2\pi n\} \tag{5}$$

Thus, the principal value of -1 is π .

Also, $\arg(\bar{z}) = \arg(z)$ if z is not a negative real number (i.e., $\varphi \neq \pi$).

Proof. Let $\varphi = \arg(z)$, then $-\pi < \varphi \leq \pi$. Since z is not a negative real number, then $\varphi \neq 0$. Also, $-\varphi \in \arg(\bar{z})$ (by reflection), also

$$-\pi < \varphi < \pi \implies -\pi < -\varphi < \pi \tag{6}$$

Thus, we conclude that $-\varphi = \arg(\bar{z})$. Therefore,

$$-\varphi = -\arg(z) = \arg(\bar{z}) \tag{7}$$

□

Exponentiation of complex numbers

Let $r = |z| = \sqrt{x^2 + y^2}$, then

$$z = r \left(\frac{x}{r} + i \frac{y}{r} \right) \tag{8}$$

Note that

$$\left(\frac{x}{r} \right)^2 + \left(\frac{y}{r} \right)^2 = 1 \tag{9}$$

Also, there is a θ such that

$$\cos \theta = \frac{x}{r}, \sin \theta = \frac{y}{r} \tag{10}$$

Thus,

$$z = r(\cos \theta + i \sin \theta) = e^{i\theta} \quad (11)$$

For $\theta \in \mathbb{R}$. In general, $z \in \mathbb{C}$ and $z \neq 0$, then

$$z = re^{i\theta}, \theta \in \arg(z) \quad (12)$$

You can also multiply complex numbers together

$$e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1 + \theta_2)} \quad (13)$$

Also, $\theta_1 \in \arg(z_1), \theta_2 \in \arg(z_2)$, then $\theta_1 + \theta_2 \in \arg(z_1 z_2)$.

Important remark

Note that $\theta_1 = \arg(z_1)$ and $\theta_2 = \arg(z_2)$, then it is not always the case that $\theta_1 + \theta_2 = \arg(z_1 z_2)$

List of important results:

1. $e^{i(\theta_1 + \theta_2)} = e^{i\theta_1} e^{i\theta_2}$
2. $e^{i(\theta_1 - \theta_2)} = e^{i\theta_1} / e^{i\theta_2}$
3. $e^{-i\theta} = 1 / e^{i\theta}$
4. $|e^{i\theta}| = 1$